

The Perihelion-Pumped Arrival-Epoch Optimum: Independent Verification and Crossing-Epoch Reconciliation

Technical memorandum for The Fermi Explorer Mission

Prepared by	Physical Superintelligence PBC
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Scope	Adjudication of the perihelion-pumped arrival-epoch optimum and the ecliptic-crossing epoch against the Fermi team's independent analysis, with the six-item measurement program that stress-tested both
Status	Final

1. Executive summary

BOTTOM LINES UP FRONT (force-ranked)

1. The Fermi team's 77,500 yr perihelion-pumped arrival optimum is correct twice over: our independent model bottoms at 77,499 yr (verified), and the adjudicated empirical basin puts both optima 16 m s^{-1} above its measured bottom, 24.156 km s^{-1} at 79,000 yr, flat to 25 m s^{-1} over 77,500 to 79,000 yr with the bottom unresolvable beyond $\pm 1,000 \text{ yr}$ (measured; Section 3).
2. The 73,012 yr impulsive-departure optimum also stands at 13.854 km s^{-1} (verified; blind re-derivation), and the two optima price different vehicles, so both are right (Section 2).
3. The 534 yr crossing-epoch gap is closed: their 79,252 yr reproduces exactly from their inputs, and the gap is +533.5 yr of inputs, distance +528.8 yr of it, plus -0.01 yr of conventions (verified; Figure 4).
4. Their input state mixes catalogs, the Akeson 2021 distance with Kervella-era proper motions, and the catalogs disagree at 5.1 sigma in parallax, so the mix matters (measured; Table 5).
5. The carried radial velocity is spectroscopic in both catalogs, the kinematic value is $-22.332 \pm 0.005 \text{ km s}^{-1}$ against the carried -22.400 , the correction moves the crossing 299 to 332 yr later, and the four catalog-convention scenarios span 79,467 to 80,118 yr on their state clocks (verified convention; measured shifts; Section 4.5).
6. The crossing design epochs cost 22.1 m s^{-1} (Akeson working state) and 27.1 m s^{-1} (Kervella) above the measured basin bottom, 0.11% of total, so the measurement endorses the Fermi design default (measured; Section 3.3).
7. End to end the crossings hold: arrival probability 0.965 (Kervella) and 0.984 (Akeson working state; 0.9858/0.9866 on the Akeson J2019.5 catalog state) against the 0.982 baseline, GTO-100 closes at both with +3.8 pts, LEO-150 still does not close, and the measured open-loop powered-phase residual is 99.6 m s^{-1} against the assumed 25 m s^{-1} per axis (simulation and measured; Section 5).
8. The joint end-to-end probability at the Kervella crossing is 0.964 at assumed execution levels and 0.606 under the measured residual, and radial velocity drives 99.1% of arrival-epoch variance (simulation; Section 6).
9. The recommended public statement is pinned to the Akeson 2021 state at its native epoch J2019.5 in the kinematic frame, with T counted from the 2029.0 departure: $T_{\text{cross}} = 79,756 \text{ yr}$ after departure, crossing date AD 81,785, program band $\pm 400\text{--}800 \text{ yr}$; the observed- γ variant gives 79,457.5 yr, AD 81,486.5 (measured; Section 7).

This memorandum closes out the two questions raised by the Fermi Explorer team’s independent perihelion-pumped (PP) arrival-epoch analysis, whether their 77,500 yr propellant optimum is correct and why their ecliptic-crossing epoch (79,252 yr) differs from ours (79,785.8 yr) by 534 yr, and it reports the six-item measurement program that stress-tested the answers: an empirical basin sweep with outlier adjudication, a radial-velocity convention audit, design-epoch dispersion and closure, powered-phase execution residuals, linearity bounds, and a joint end-to-end Monte Carlo. All questions are settled above. Every number in this memorandum traces to a machine artifact or to the Fermi team’s cited files, and every claim carries its evidence grade in the sentence that states it. Neither implementation was tuned toward the other at any point. An independent adversarial verification pass re-ran every analysis script to completion, recomputed the headline values through different call paths, and confirmed every verdict (Appendix A).

Two subsidiary findings complete the picture. The $0.04 \text{ km s}^{-1} v_\infty$ offset between the two epoch tables has the same single cause as the crossing gap: the raw unshaved v_∞ at 73,000 yr is $23.7145 \text{ km s}^{-1}$ on their inputs and $23.8087 \text{ km s}^{-1}$ on ours, convention-independent to 0.0001 km s^{-1} , and distance drives the difference (measured; Section 4.7). And the Fermi team’s own error budget anticipated the main result: their error-budget table lists “Astrometry inputs (sub-0.5% differences move the crossing itself ~ 500 yr)”, which is exactly the effect we measure. Their numerical discipline, their dt/8 convergence checks, and their custody-gated acceptance criteria all held up under our audit (Section 3).

2. The two-optima structure: why both numbers are correct

The arrival epoch T fixes the required departure asymptote: the heliocentric cruise speed $v_\infty(T)$ and its ecliptic latitude $\beta(T)$, both set by the Alpha Centauri AB barycentric astrometry with the 2,600 AU miss allowance credited. Three different objectives place their minima at three different epochs.

- **Cruise speed.** $v_\infty(T)$ alone bottoms at 57,854 yr at 23.173 km s^{-1} (measured; 1 yr grid over 50,000 to 100,000 yr). Over the 70,000 to 82,000 yr window relevant here, $v_\infty(T)$ rises monotonically (measured on a 49-point grid).
- **Impulsive departure.** The single-impulse Oberth departure from a 400 km LEO pays $\sqrt{v_{\text{esc}}^2 + v_\infty^2} - v_{\text{circ}}$, which trades the rising $v_\infty(T)$ against the shrinking geometric tilt. Its minimum is 73,012.17 yr at $13.8539 \text{ km s}^{-1}$ (verified; blind re-derivation agreed with the heritage value 73,012.04 yr to 0.2 yr).
- **Perihelion-pumped campaign.** The PP vehicle acquires tilt by steering thrust out of plane during the pump campaign. That cost is a separate additive term that falls quadratically to zero where the asymptote crosses the ecliptic (measured; decay exponent 1.92, Section 3). Adding it to the shaved $v_\infty(T)$ moves the minimum to 77,499 yr in our model (measured) and 77,500 yr in their direct simulation (their table; verified against our model to 1 yr), and the measured basin of Section 3.3 places both inside its flat bottom.

The 73,012 yr and 77,500 yr numbers therefore answer different questions. The impulsive optimum prices a chemical departure stage. The PP optimum prices the pumped vehicle’s own propellant. The Fermi team’s document scopes their analysis to the PP vehicle explicitly, and within that scope their epoch table is correct (verified; Section 3). Their design default is the ecliptic crossing itself, 33 m s^{-1} above their own basin bottom and 22 to 27 m s^{-1} above our measured basin bottom (measured; Section 3.3), chosen because it is the one epoch fixed by geometry alone. That design logic survives our adjudication and our measurement program: the crossing epoch is invariant under every pricing-model revision, its residual uncertainty is astrometric (Section 4), and its end-to-end dispersion and vehicle closure hold at the crossing (Section 5).

3. Independent verification of the PP optimum

3.1 Measured tilt-acquisition costs

We measured the 3-D tilt-acquisition cost directly with the certified propagator at four arrival epochs, holding everything else fixed. Each row targets the exact kinematic aim (v_∞, β) of its epoch, runs the certified 3-D pump search under the heritage acceptance gates plus a direction gate $|\beta_{\text{achieved}} - \beta_{\text{target}}| \leq 0.04^\circ$, and subtracts a matched planar pump run at the same v_∞ target. The carried value is the maximum of the gate-passing

Table 1. Measured 3-D tilt-acquisition cost versus asymptote ecliptic latitude. Carried 3-D and matched-planar totals are best-found values under identical gates and seeds (seed base 20260803); the increment column is their difference, computed before rounding. The zero-tilt row is the planar-collapse control: the 3-D search at $\beta_{\text{target}} = 0$ lands 0.015 km s^{-1} above its own planar embedding, which bounds the method’s residual.

Arrival T (yr)	$ \beta_{\text{target}} $ (deg)	Carried 3-D (km s^{-1})	Matched planar (km s^{-1})	Tilt cost (km s^{-1})
73,012	2.48	24.563	24.006	0.556
75,000	1.70	24.345	24.052	0.293
77,500	0.78	24.173	24.111	0.062
79,786	0.00	24.200	24.200	0.000

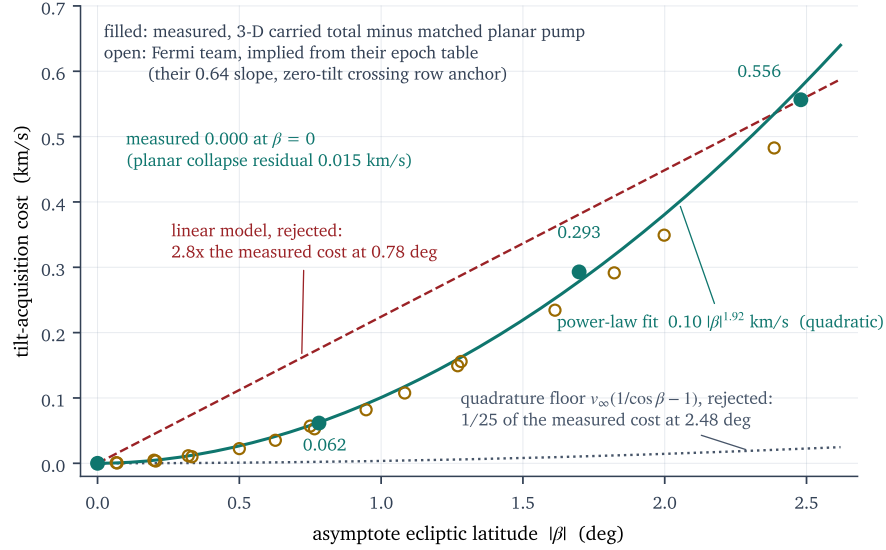


Figure 1. Tilt-acquisition cost versus asymptote ecliptic latitude. Filled points: our measured 3-D carried totals minus the matched planar pump (Table 1). Open points: the Fermi team’s implied per-epoch tilt costs from their published table. The quadratic fit $0.10 |\beta|^{1.92} \text{ km s}^{-1}$ (measured) passes through both data sets. The linear model and the quadrature floor, both fixed a priori, are rejected at ratios 2.8 and 25 (measured).

$\times 1.0/\times 0.5/\times 0.25$ time-step ladder. The 3-D totals are best-found values from a seeded stochastic search with per-row scatter of $\pm 0.2 \text{ km s}^{-1}$; the differences in Table 1 are the quantity the model uses.

A power-law fit to the measured increments gives $0.10 |\beta|^{1.92} \text{ km s}^{-1}$ with β in degrees (measured; amplitude 0.1007, exponent 1.92, quadratic within the scatter). Two alternative models fixed a priori are rejected by the same data: a linear model anchored at the 2.48° point overprices the 0.78° epoch by a factor of 2.8, and the geometric quadrature floor $v_\infty(1/\cos \beta - 1)$ underprices the 2.48° point by a factor of 25 (measured; Figure 1). The quadratic law was asserted from the single heritage anchor before the multi-epoch measurement and was not adjusted afterward.

3.2 The epoch curve and the 1 yr agreement

Our tilt-inclusive model is the shaved kinematic $v_\infty(T)$ plus the quadratic anchor term $0.58 (\beta(T)/2.48^\circ)^2 \text{ km s}^{-1}$, with the 0.58 km s^{-1} anchor taken from the heritage 3-D measurement before this analysis began. Its minimum is 77,499 yr (measured). The Fermi team’s direct simulation, an unrelated implementation with its own thermal power curve, steering law, and step control, bottoms at 77,500 yr on a 500 yr grid with a basin flat to 7 m s^{-1}

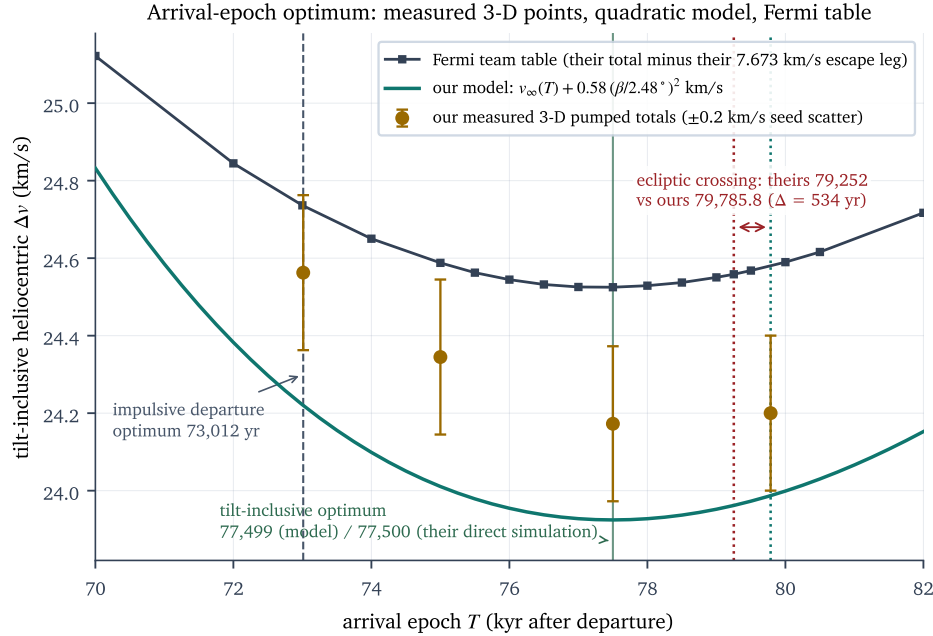


Figure 2. Arrival-epoch curves. Dark curve: the Fermi team’s published tilt-inclusive totals with their 7.673 km s^{-1} escape leg removed. Teal curve: our model, $v_{\infty}^{\text{shaved}}(T) + 0.58 (\beta/2.48^{\circ})^2 \text{ km s}^{-1}$ (measured components). Points with bars: our measured 3-D pumped totals with the $\pm 0.2 \text{ km s}^{-1}$ best-found seed scatter. The two curves sit at different absolute levels because their direct simulation prices the full campaign under its own conventions while our model prices kinematics plus the tilt anchor; the adjudicated quantity is each curve’s basin location, which the level offset does not move (their record states the convention deltas are epoch-smooth). The two minima, 77,499 yr (ours, measured) and 77,500 yr (theirs), agree to 1 yr, and both sit 16 m s^{-1} above the measured basin bottom of Figure 3. Vertical lines mark the impulsive optimum at 73,012 yr and the two crossing epochs whose 534 yr gap Section 4 decomposes; Figure 4 resolves that gap bar by bar.

over 76,500 to 78,000 yr (their table; verified against our model to 1 yr). Figure 2 overlays the two curves and our four measured 3-D points. No parameter in either implementation was tuned toward the other. Both minima are model results; the direct 15-epoch measurement of Section 3.3 locates the physical basin around them. Their own cross-checks strengthen the agreement. Their independent 3-D re-integration of the tilt cost, written from a docstring specification with its own stepping, gives 610 m s^{-1} at the 2.48° point against their engine’s 606 m s^{-1} and our measured 578 m s^{-1} : three implementations inside a 5.5% spread (verified from their audit record and ours). Their step-size discipline is equally concrete: a $dt/8$ re-integration of three checked rows moves the totals by at most 4.1 m s^{-1} (75,000 yr: -3.2 ; 77,500 yr: -3.5 ; crossing: -4.1 m s^{-1}) against their 40 m s^{-1} audit gate (their record; verified against their published cross-check list). Their per-row acceptance discipline is explicit as well: every epoch row must reach the target v_{∞} , land the asymptote latitude within 0.06° of $-\beta$, and keep stellar custody at or below 15 yr (their record). Their adversarial audit history is documented and credible: they found and corrected a one-sided miss-allowance convention that overpriced tilted epochs by up to 100 m s^{-1} , and they retracted an earlier corroboration claim that rested on a seed-scatter outlier in our planar table.

3.3 The measured basin: swept, adjudicated, flat

The model agreement above is necessary evidence; the definitive evidence is a direct empirical re-trace of the tilt-inclusive 3-D pumped basin. The sweep covers epochs 74,000 to 81,000 yr in 500 yr steps with five seeds per epoch (seed base 20260804, explicit per-run offsets) and imports the bounded 3-D search recipe verbatim

Table 2. Basin-sweep anchor gates (verified; from the completed sweep artifact, re-checked by the adjudication run). The collapse gate is 0.05 km s^{-1} , declared in-script against the prior measured 0.0149 km s^{-1} ; it is a declared loose gate with its prior value referenced.

Anchor	Quantity	Value (km s^{-1})	Result
A	Heritage planar carried Δv	23.984510138158	pass (bit-level)
B	Heritage certified 3-D carried Δv	24.562696	pass (bit-level)
C	77,500 yr winner re-ladder	24.172672236	pass (exact vs prior artifact)
D	Crossing-winner planar collapse	+0.014893	pass (gate 0.05, declared)

from the certified heritage code. All four anchor gates pass (Table 2). The sweep completed in 1,715 s against its 2,100 s budget. All 75 cells are feasible at 5/5 seeds (measured).

Two cells show the single-seed outlier signature seen before in this project: seed 20260918 at 79,500 yr found a 23.830 km s^{-1} schedule 0.33 km s^{-1} below both neighbors, and seed 20260938 at 80,500 yr found 24.114 km s^{-1} , 0.07 to 0.12 km s^{-1} below both neighbors, each from a remote corner of the search box. An adjudication run re-ran the 79,500 cell with five fresh seeds and the 80,500 cell with three, spot re-ran 78,500 and 79,000, and re-laddered both outlier winners on an extended dt ladder down to $\times 0.0625$. The verdict is clean (measured). Zero of eight fresh seeds reproduce either mode; all eight converge to the family values, 24.186090 and $24.228154 \text{ km s}^{-1}$. Both spot re-runs reproduce their sweep values exactly. Both outlier winners pass all gates on the extended ladder; the 79,500 winner drifts $+12.5 \text{ m s}^{-1}$ at the finer rungs and the 80,500 winner drifts 0.0 m s^{-1} . Each outlier run is therefore a valid upper bound at its own epoch, found by one seed in ten (79,500) or one in eight (80,500), and both are excluded from the envelope because a mode that rare distorts cross-epoch comparison under a fixed search budget. The exclusion is conservative for the design conclusion: a rare deeper schedule family near 79,500 to 80,500 yr moves the basin bottom toward the crossing epochs.

The adjudicated envelope (Figure 3) bottoms at 24.156 km s^{-1} at 79,000 yr with zero within-cell seed scatter at all 15 cells (measured; every value a best-found upper bound). Flatness: 78,500 to 79,000 yr lie within 10 m s^{-1} of the bottom, 77,500 to 79,000 within 25 m s^{-1} , and 77,000 to 80,000 within 50 m s^{-1} . The basin is too flat to localize its bottom beyond $\pm 1,000$ yr: the 25 m s^{-1} band spans 1,500 yr, and the quadratic fits put the vertex at 78,747 yr (five cells around the bottom) or 78,589 yr (full window) with rms residuals of 7.7 and 7.3 m s^{-1} . That flatness is the finding. The 77,499 yr model fit and Fermi's 77,500 yr direct optimum each sit 16.3 m s^{-1} above the measured bottom, inside the 25 m s^{-1} band, so the verified 1 yr model agreement and the measured basin tell one consistent story; Fermi's own table holds 76,500 to 78,000 yr within 7 m s^{-1} of their bottom, which matches a basin this flat. The crossing design epochs cost 22.1 m s^{-1} (Akeson-based working state, Table 5, 79,372 yr) and 27.1 m s^{-1} (Kervella, 79,786 yr) above the measured bottom, both inside the 50 m s^{-1} band. Choosing the crossing as the design epoch gives up under 30 m s^{-1} of Δv , 0.11% of the 24.16 km s^{-1} total, so the crossing design epoch costs almost nothing and the measurement strengthens the recommendation of Section 7. The joint Monte Carlo of Section 6 carries a surrogate-roughness term $\sigma_{\text{sur}} = 0.0091 \text{ km s}^{-1}$; the measured quadratic-detrend rms of the adjudicated envelope is 0.0073 km s^{-1} , so the surrogate over-covers the measured roughness and stands as conservative (measured).

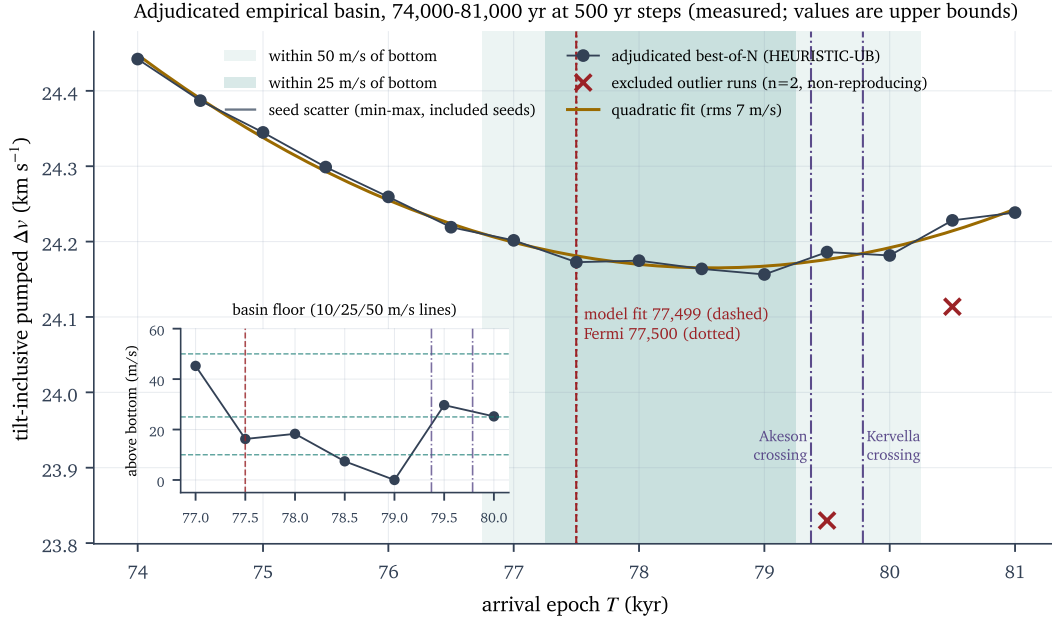


Figure 3. Adjudicated empirical basin (measured; every point a best-of-N upper-bound envelope value). Points: adjudicated best per epoch with min-max seed whiskers (zero spread at every cell); red crosses: the two excluded single-seed outlier runs, with zero of eight fresh seeds reproducing them; amber curve: full-window quadratic fit (rms 7 m s⁻¹); shaded spans: epochs within 25 and 50 m s⁻¹ of the bottom; dashed and dotted red lines: the model fit 77,499 and Fermi 77,500 yr, each 16 m s⁻¹ above the measured bottom; dash-dot purple lines: the Akeson working-state (79,372) and Kervella (79,786) crossing design epochs, 22 and 27 m s⁻¹ above the bottom. Inset: depth above the bottom in m s⁻¹ with the 10/25/50 m s⁻¹ thresholds marked.

4. The crossing-epoch reconciliation

Both teams define the ecliptic-crossing epoch as the arrival time T at which the required departure asymptote has zero ecliptic latitude. Ours is 79,785.8 yr, held by three independent methods within 0.05 yr (verified). Theirs is 79,252 yr. The 534 yr gap is the one numerical disagreement between the two analyses, and this section closes it.

4.1 Their implementation, dissected

We read the Fermi team's public source: `astro.py`, `constants.py`, `intercept.py`, and `pump_schedule.py` in `fermi_sim`, plus `tools/sim_pp_arrival.py`. Their crossing condition is $T = -z_0/v_z$ applied to the ecliptic-frame straight-line stellar state. Ours is the root of $\beta(T) = 0$. For a linear state these are algebraically identical, and we confirmed the identity numerically to 0.01 yr (verified). Their code is clean: the rotation to ecliptic coordinates, the proper-motion scaling, and the radial-velocity projection all match textbook form. The two implementations differ only in their input astrometry and in trailing digits of shared constants.

4.2 Reproduction: their number, from their inputs, in our code

We reran a clean reimplement on both input sets under both convention sets (obliquity digits, light-year value, proper-motion conversion digits). The full 2×2 matrix is Table 3. The reproduction is a full success with no residual to characterize: their inputs give 79,252.25 yr under either convention set, inside the integer rounding of their published 79,252 yr (verified). Our inputs give 79,785.75 yr under either convention set, matching our reference to 0.05 yr (verified). No further information from the Fermi team is needed.

Table 3. Reproduction matrix (verified; clean reimplement, rerun 2026-08-03). Crossing epoch in years and raw unshaved v_∞ at 73,000 yr in km s^{-1} , for each combination of input set and convention set. Conventions move the crossing by 0.01 yr and v_∞ by under 0.0001 km s^{-1} ; inputs move them by 533.5 yr and 0.094 km s^{-1} .

Combination	T_{cross} (yr)	Raw v_∞ at 73 kyr (km s^{-1})
Fermi inputs, Fermi conventions	79,252.25	23.7145
Fermi inputs, our conventions	79,252.24	23.7145
Our inputs, Fermi conventions	79,785.75	23.8087
Our inputs, our conventions	79,785.75	23.8087

Table 4. One-at-a-time attribution of the 533.5 yr input effect, swapping each Fermi input to ours (measured). Distance carries 99% of the gap. The proper-motion terms partially cancel. The radial velocity is identical in both states, the declinations differ in the fourth decimal, and both rows contribute zero at the displayed precision. The single-swap sum leaves a +0.1 yr interaction residual against the joint effect, and an independent fresh-script re-derivation in the opposite swap direction returns -533.6 yr.

Input	Fermi value	Our value	Shift (yr)
Distance	4.344 ly	4.36522 ly (747.17 mas)	+528.8
μ_δ	694.0 mas yr $^{-1}$	693.8 mas yr $^{-1}$	+6.0
μ_{α^*}	$-3620.0 \text{ mas yr}^{-1}$	$-3619.9 \text{ mas yr}^{-1}$	-1.4
Right ascension	219.9021°	219.90206°	+0.03
Declination	-60.8340°	-60.83399°	0.00
Radial velocity	identical	identical	0.00
Sum of one-at-a-time swaps			+533.4
Interaction residual			+0.1
Joint input effect (Table 3)			+533.5
Reverse-direction check (our state to theirs)			-533.6

4.3 Decomposition: input effect, convention effect

The published gap is $79,785.75 - 79,252.00 = 533.75$ yr. Of this, 0.25 yr is their integer rounding; the input effect is +533.50 yr at their conventions (+533.51 at ours) and the convention effect is -0.01 yr at their inputs (0.00 at ours), so each decomposition path closes the printed bridge exactly (measured; Table 3). A one-at-a-time swap of each input from their value to ours attributes the input effect in Table 4, and Figure 4 draws the same decomposition as a cumulative bridge from their reproduced value to ours (measured).

4.4 The catalog question: a 5-sigma parallax tension

Neither state is wrong on its face; the two states draw on different catalogs, and Table 5 sets them side by side. Their 4.344 ly distance equals the Akeson 2021 parallax (750.81 ± 0.38 mas), the more recent and more precise measurement, while their proper motions are Kervella-era. Ours is Kervella 2016 (747.17 ± 0.61 mas) end to end. Their state mixes catalogs; ours is internally consistent but older.

The mix matters because the two catalogs disagree. The parallaxes differ by 3.64 mas against a combined one-sigma of 0.72 mas, a 5.1-sigma tension (computed from the published parallax sigmas in Table 5). The main feasibility report carries this same Kervella–Akeson tension as a labeled systematic in its dispersion analysis, and this memorandum inherits that treatment. Epoch consistency is a second, independent lever. Attaching Akeson’s J2019.5 observables to a J2000-family position and velocity basis moves the crossing earlier by 31.6 yr on the state clock (24.5 yr of velocity-basis rotation plus 7.1 yr of position advance) and by 51.1 yr on the absolute date under the J2000-origin reading (measured). The epoch-consistent Akeson state at J2019.5 gives $T_{\text{cross}} = 79,467.0$ yr from J2019.5 in the spectroscopic V_0 frame (measured); the working state of Table 5 sits on the J2000-family basis with the adopted proper motions, so its 79,371.9 yr differs by the epoch basis and the observable deltas together; the frame itself is the next subsection’s subject.

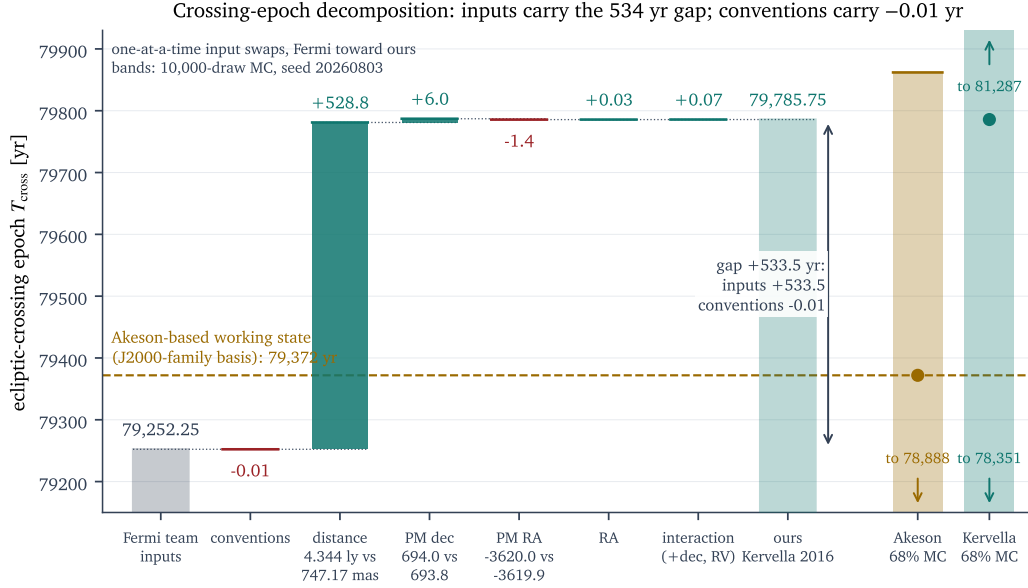


Figure 4. Waterfall decomposition of the crossing-epoch gap (measured). The bridge starts at 79,252.25 yr, our exact reproduction of the Fermi state, published by them as 79,252 yr under integer rounding, and ends at our 79,785.75 yr. Bars in order: conventions -0.01 yr, distance +528.8 yr, μ_δ +6.0 yr, $\mu_{\alpha*}$ -1.4 yr, right ascension +0.03 yr, interaction residual +0.07 yr; bar labels round independently, and the drawn bars close the bridge exactly at 79,785.75 yr. Distance carries 99% of the gap. The dashed line marks the Akeson-based working state at 79,371.9 yr (J2000-family basis; the catalog-epoch J2019.5 state gives 79,467.0 yr, Section 4.4). The shaded columns at right give the 68% Monte Carlo intervals under each catalog: [78,888, 79,862] yr for Akeson 2021 and [78,351, 81,287] yr for Kervella 2016, with arrows marking extents beyond the plotted range.

4.5 The radial-velocity convention audit

The crossing epoch moves $4,864$ yr per km s^{-1} of radial velocity (measured), so the radial-velocity convention is a candidate systematic larger than most inputs. The published catalog values are spectroscopic: gravitational redshift and convective blueshift stay uncorrected. Trajectory work needs the kinematic value, the true line-of-sight motion. We audited the carried value against the primary sources. The verdict is definite (verified against re-fetched source text): the carried -22.4 km s^{-1} is the spectroscopic Kervella 2016 orbit gamma (-22.393 km s^{-1} ; fit to all available RV measurements, with CES and HARPS shown in their residuals), and Akeson 2021's $V_0 = -22.3796$ km s^{-1} is the same convention. The kinematic correction is smaller than the naive sum of the published $+575.3$ m s^{-1} (~ 0.58 km s^{-1}) gravitational redshift and a -0.3 km s^{-1} convective blueshift. Alpha Centauri A is a solar twin measured against a solar-referenced zero point, so the correction collapses to removing the differential gravitational redshift, published as $\Delta\nu_{\text{GRS}} = -61.4 \pm 2.7$ m s^{-1} (Kervella, Thévenin & Lovis 2017, A&A 598, L7); the applied zero-point correction is $+61.4$ m s^{-1} , a hidden one-sided systematic of $+0.061$ km s^{-1} . The kinematic Kervella value is -22.332 km s^{-1} . Table 6 prices every scenario; Figure 6a draws the ladder.

Three consequences follow (measured). First, the kinematic-frame central crossing is 332 yr later on the Kervella base and 298.9 yr later on the epoch-consistent Akeson base (349 on the working state), 0.23 of the 0.30 km s^{-1} carried working sigma (adopted allowance) and 0.6 of the 0.10 allowance carried with the Akeson-based state, so no published band is invalidated and the central value carries a known one-sided systematic. Second, the correction cannot close the ours-minus-Fermi gap: if both teams use the same catalog RV the relative gap is convention-invariant, and the correction moves both absolute values in the same direction. Third, our own constants module (not the Fermi repository) carries an inconsistency worth one comment line: it holds the kinematic -22.204 km s^{-1} for Proxima (Kervella, Thévenin & Lovis 2017, A&A 598, L7) beside the

Table 5. Input comparison: the Fermi published state, the two catalogs (Kervella 2016, A&A 594, A107, catalog epoch J1991.25; Akeson 2021, AJ 162, 14, native epoch J2019.5), and the Akeson-based working state that the dispersion and joint Monte Carlo machinery runs on. In the Kervella 2016 column the parallax, proper motions, and gamma are verbatim catalog values; the position (219.90206, -60.83399) is a project constant on a J2000-family basis and is not a K16 catalog value (the K16 catalog barycenter is published at J1991.25). In the working-state column the proper motions ($-3644.4, +702.8$) and their $\pm 1.5 \text{ mas yr}^{-1}$ sigmas are adopted working values that appear in no catalog (Akeson’s published sigmas are $0.42/0.17$); the -22.39 km s^{-1} is the K16/PB16 gamma rounded to two decimals (Akeson’s own V_0 rounds to -22.38); the $0.30/0.10 \text{ km s}^{-1}$ sigmas are working sigmas, adopted absolute-accuracy allowances that appear in neither catalog. T_{cross} for the Akeson column counts from its J2019.5 state (spectroscopic V_0 ; kinematic 79,765.9); the other columns count from their own state clocks (measured).

Quantity	Fermi state	Kervella 2016	Akeson 2021 (J2019.5)	Working state (adopted)
Parallax (mas)	(via distance)	747.17 ± 0.61	750.81 ± 0.38	750.81 ± 0.38
Distance (ly)	4.344	4.36522	4.344	4.344
$\mu_{\alpha*}$ (mas yr $^{-1}$)	-3620.0	-3619.9 ± 3.9	-3639.95 ± 0.42	-3644.4 (adopted)
μ_{δ} (mas yr $^{-1}$)	$+694.0$	$+693.8 \pm 3.9$	$+700.40 \pm 0.17$	$+702.8$ (adopted)
Radial velocity (km s $^{-1}$)	-22.4	-22.3930 ± 0.0043	-22.3796 ± 0.0020	-22.39 (allowance 0.10)
Right ascension (deg)	219.9021	219.90206	219.8589	219.90206
Declination (deg)	-60.834	-60.83399	-60.8316	-60.83399
T_{cross} from state (yr)	79,252.25	79,785.75	79,467.0	79,371.9

Table 6. RV-convention scenarios and their crossing epochs (measured; exact crossing solve. Linear cross-checks agree within 2 yr on the single-state RV-swap rows; the bounding-case row departs from linearity by 19 yr, and the two J2019.5-state rows’ ΔT combine the state change with the RV change). The first block runs on the K16 astrometric base, the second on the A21 base; the working-state row runs on the Akeson-based working state of Table 5 at its adopted precision -22.39 , while the epoch-consistent rows run on the Akeson J2019.5 state with T from J2019.5. The four catalog-convention scenarios span 79,467 to 80,118 yr. The final row is a bounding case only, contradicted by the Kervella 2017 frame analysis.

Scenario	Convention	RV (km s $^{-1}$)	T_{cross} (yr)	ΔT vs block baseline (yr)
Carried K16 (constants)	spectroscopic	-22.400	79,785.75	0.0
K16 published gamma	spectroscopic	-22.393	79,819.81	+34.1
Pourbaix–Boffin 2016	spectroscopic	-22.390	79,834.42	+48.7
K17 kinematic	kinematic	-22.332	80,117.90	+332.2
A21 working state (constants)	spectroscopic	-22.39	79,371.93	0.0
A21 published V_0 (J2019.5 state)	spectroscopic	-22.3796	79,467.02	+95.1
A21 kinematic (J2019.5 state)	kinematic	-22.3182	79,765.87	+393.9
Naive uncorrected bound	(unsupported)	-22.657	78,554.90	$-1,230.8$

spectroscopic -22.40 km s^{-1} for AB, two conventions in one file. The Fermi constants file carries no Proxima entry and no stellar radial velocity; their AB value (-22.4 km s^{-1} , in `astro.py`) is spectroscopic throughout. If the zero-point argument were rejected wholesale, the bounding range on the correction is $[-0.27, +0.07] \text{ km s}^{-1}$, which maps to $[-1.3, +0.34] \text{ kyr}$, with the literature-supported value at $+0.061 \text{ km s}^{-1}$ (bounds; the -1.3 kyr endpoint is contradicted by Kervella 2017).

Table 7. Sigma positions of the three crossing-epoch values under each working-sigma band (measured; linear one-sigma propagation, band centers at each state, spectroscopic frame). Every entry sits inside the 68% interval of both bands, so no pair of values is in statistical tension. The kinematic correction shifts the band centers later by 332 yr (Kervella) and 349 yr (the Akeson-based working-state band drawn here; 299 yr applies on the epoch-consistent center) and does not change the relative positions.

Crossing-epoch value	Kervella band (79,786 $\pm$ 1,468 yr)	Akeson band (79,372 $\pm$ 489 yr)
Fermi published, 79,252 yr	-0.36σ	-0.25σ
Akeson working state, 79,372 yr	-0.28σ	0.00 σ (center)
Ours (Kervella), 79,786 yr	0.00 σ (center)	$+0.85\sigma$

4.6 Sensitivity ranking and the astrometric band

Linear sensitivities of the crossing epoch (measured, central differences on the Kervella state): radial velocity $+4,864$ yr per km s^{-1} , parallax -146 yr per mas, $\mu_\delta -31$ yr per mas yr^{-1} , $\mu_{\alpha*} -14$ yr per mas yr^{-1} . Weighted by the carried working sigmas (adopted absolute-accuracy allowances; the published catalog sigmas are smaller), radial velocity dominates: the 0.30 km s^{-1} allowance carried with the Kervella-based state alone contributes 1,459 yr (Figure 5a).

Propagating the full diagonal covariance (assumed diagonal; the constants file carries no cross-terms) gives the crossing-epoch bands (measured; 10,000-draw Monte Carlo, seed 20260803, linear check within 0.5%):

- **Kervella 2016:** 79,786 $\pm$ 1,468 yr linear, Monte Carlo sigma 1,462 yr, 68% interval [78,351, 81,287] yr, 95% interval [77,029, 82,755] yr. The Fermi value sits at -0.36σ , inside the 68% interval.
- **Akeson 2021:** 79,372 $\pm$ 489 yr, 68% interval [78,888, 79,862] yr, 95% interval [78,434, 80,332] yr. The Fermi value sits at -0.25σ and ours at $+0.85\sigma$, both inside the 68% interval. This is a working-sigma band on the working-state center: the 0.10 km s^{-1} RV allowance dominates the width, so the width is insensitive to the adopted $\pm 1.5 \text{ mas yr}^{-1}$ sigmas (on the Akeson-centered sensitivities the published 0.42/0.17 move the linear sigma from 489 to 487 yr), and the epoch-consistent center is 95 yr later (computed from the measured sensitivities).

Both bands are computed in the spectroscopic frame that both catalogs publish. The kinematic correction of Section 4.5 shifts each band center later by 332 yr (Kervella) or 299 yr (epoch-consistent Akeson base; 349 on the working state) and leaves the widths unchanged: the 2.7 m s^{-1} correction uncertainty contributes 13 yr, negligible against either sigma (computed from the measured sensitivities). The sigma positions in Table 7 therefore hold in either frame up to a common shift.

The 534 yr gap is therefore not statistically significant under either catalog (measured; Figure 5b). Table 7 places all three crossing-epoch values inside both bands: every entry sits inside its 68% interval. Closing the gap entirely within a single Kervella parameter would need a 0.37σ shift in radial velocity, a routine excursion; the next-cheapest single-parameter closure costs 4.5σ in μ_δ (measured).

The radial-velocity payoff quoted in Section 7 traces to these sensitivities: a 0.03 km s^{-1} radial-velocity sigma contributes $0.03 \times 4,864 = 146$ yr, the non-radial-velocity Akeson terms add 74 yr in quadrature, and the combined band is 164 yr, quoted conservatively as ± 200 yr (computed from the measured linear sensitivities).

4.7 The ν_∞ offset: same cause, same verdict

The Fermi table runs 0.04 km s^{-1} below ours at matched epochs (23.60 against 23.64 km s^{-1} at 73,000 yr). The raw unshaved ν_∞ at 73,000 yr is $23.7145 \text{ km s}^{-1}$ on their inputs and $23.8087 \text{ km s}^{-1}$ on ours, and the value is convention-independent to 0.0001 km s^{-1} (measured; Table 3). The one-at-a-time attribution assigns $-0.0951 \text{ km s}^{-1}$ to distance, $+0.0006 \text{ km s}^{-1}$ to $\mu_{\alpha*}$, and $+0.0002 \text{ km s}^{-1}$ to μ_δ (measured); distance drives more than the whole gap and the proper-motion terms partially offset it. The raw 0.094 km s^{-1} gap compresses to the 0.04 km s^{-1} table-level gap after each side applies its own miss-allowance shave. Neither implementation is wrong; the offset is the same distance difference seen through the velocity projection.

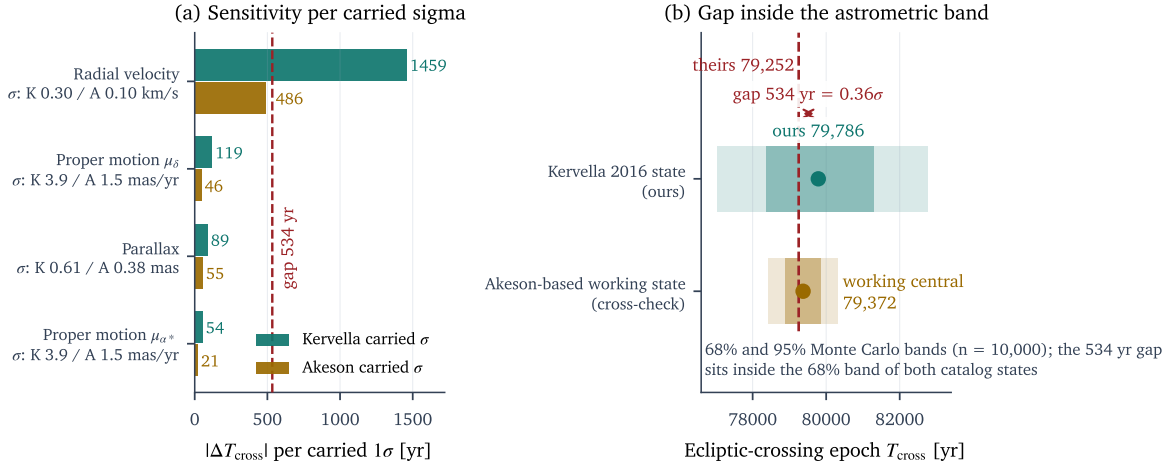


Figure 5. The crossing-epoch gap in its astrometric context. (a) Sensitivity of T_{cross} per one-sigma shift of the carried working sigmas under both catalog states (measured; the RV sigmas are the adopted $0.30/0.10 \text{ km s}^{-1}$ allowances, and the Akeson μ bars use the adopted $\pm 1.5 \text{ mas yr}^{-1}$; the published $0.42/0.17 \text{ mas yr}^{-1}$ sigmas give bars of ≈ 6 and 5 yr); radial velocity dominates both. (b) The 68% and 95% Monte Carlo bands (10,000 draws, seed 20260803) for the Kervella 2016 and Akeson 2021 states, spectroscopic frame; the 534 yr gap between our 79,786 yr and the Fermi 79,252 yr sits inside the 68% band of each. The Akeson band drawn here is centered on the Akeson-based working state (79,372 yr; the epoch-consistent center is 95 yr later); the recommended public statement (Akeson J2019.5 state, kinematic frame, T from departure 2029.0, program band $\pm 400\text{--}800 \text{ yr}$) is in Section 7.

5. Design-epoch consequences

5.1 Dispersion and closure at the crossing design epochs

The crossing recommendation deserves an end-to-end price, and the certified dispersion pipeline supplies it: 20,000 draws per case, identical error budget to the published baseline (pointing 0.10° per axis, magnitude 50 m s^{-1} , residual 25 m s^{-1} per axis, all engineering assumptions; carried ephemeris sigmas (Table 5); 2,600 AU success radius). Table 8 gives the arrival probabilities and Table 9 the vehicle closure; Figure 6b draws the probability ladder.

The findings hold no surprises (simulation). The crossing design epochs cost about 1.6 points of arrival probability against the 73,012 yr baseline, and on the Akeson working state the crossing beats the baseline by +0.23 points. GTO-100 closes at both crossings and gains +0.69/+0.71 pts over the 73k tilt-inclusive case (24.56 km s^{-1}). The LEO-150 verdict does not change: it still fails to close at either crossing. Its shortfall shrinks from -0.74 pts at 73,012 yr to $-0.12/-0.10$ pts, close enough that a 0.06 (Akeson working) to 0.08 km s^{-1} (Kervella) Δv saving would flip it, and we do not claim the flip. On the Akeson J2019.5 catalog state (same pipeline, closest-approach scoring, $n = 20,000$, seed 20260806), the central dispersion probability is 0.9858 in the kinematic frame and 0.9866 in the observed- γ variant; the working state gives 0.9846 at the same seed (Table 8's 0.98385 is the same working state at seed 20260805, a within-noise seed difference), and a cross-catalog diagnostic that aims on the working-state proper motions against the J2019.5 state prices that input difference at $P = 0.9660$ with median miss 1,193 AU (simulation).

Table 8. Arrival dispersion at the design epochs (simulation; $n = 20,000$ per row, seeds stated, Wilson 95% intervals). The baseline row reproduces the published seed-42 anchor exactly. The last row is an interpretation control: the P deficit follows the catalog (aim-state error), and the epoch itself is benign.

Case	T (yr)	Seed	$\hat{P}$	Wilson 95%	Median miss (AU)
Baseline, Kervella, 73k	73,012	42	0.98155	[0.9796, 0.9833]	1,076
Kervella crossing	79,786	20260804	0.96515	[0.9625, 0.9676]	1,173
Akeson crossing (working state)	79,372	20260805	0.98385	[0.9820, 0.9855]	1,059
Kervella catalog at Akeson epoch	79,372	20260806	0.96695	[0.9644, 0.9693]	1,171

Table 9. Vehicle closure at the crossing epochs (measured slopes, best-found Δv rows). Required Δv is a linear interpolation of the 12-yr pumped column between the $T = 79,000$ (24.156 km s^{-1}) and $T = 80,000$ (24.187 km s^{-1}) best-found rows, an upper bound, so margins are lower bounds; residual tilt at these epochs is $\leq 0.008 \text{ km s}^{-1}$ by the measured power law. Margins extrapolate the bracket-end margins (+6.15 pts at 22.9 km s^{-1} , +4.15 at 24.0 for GTO-100; +1.96/+0.17 for LEO-150) at the measured slopes -1.818 and -1.627 pts per km s^{-1} .

Architecture	Kervella 79,786		Akeson working 79,372		Verdict
	Δv (km s^{-1})	Margin (pts)	Δv (km s^{-1})	Margin (pts)	
GTO-100	24.180	+3.82	24.168	+3.85	closes (both)
LEO-spiral-150	24.180	-0.12	24.168	-0.10	does not close

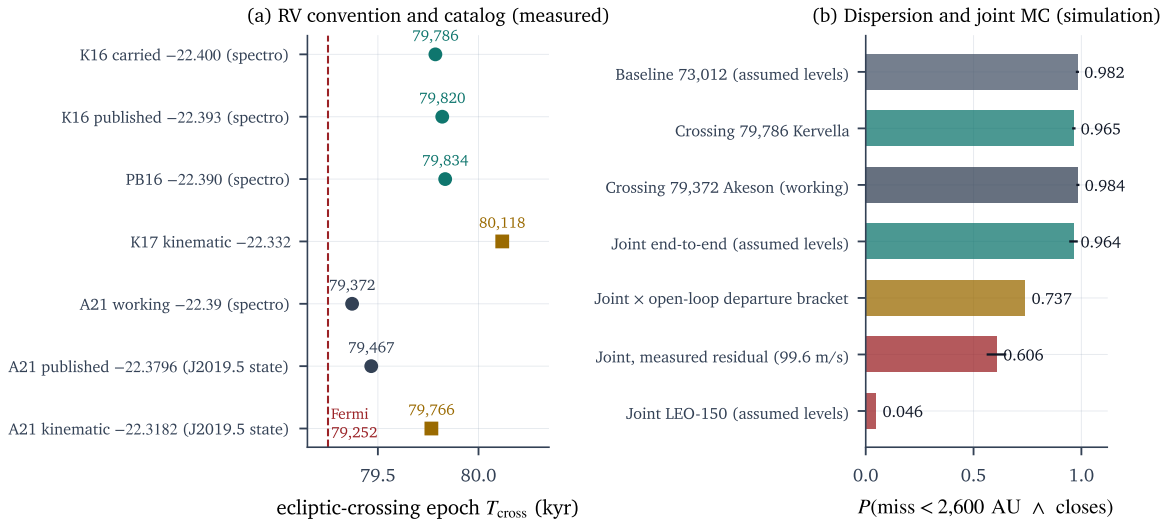


Figure 6. Measurement-program summary panel, drawn from program artifacts under anchor gates. (a) Crossing epoch under each RV convention and catalog (Table 6; measured); squares mark kinematic-frame values, the dashed line the Fermi published value; the two J2019.5-state rows carry the catalog-state values, and the four catalog-convention scenarios span 79,467 to 80,118 yr. (b) Probability ladder (simulation): dispersion at the three design epochs with Wilson 95% intervals (the Akeson bar is the working state; the J2019.5-state values 0.9858/0.9866 are in Section 6), then the joint end-to-end Monte Carlo (Section 6); whiskers on joint bars are the 2.5–97.5% range of the per-draw dispersion probability, and the open-loop departure bracket and measured-residual rows are conservative upper-bound stresses.

5.2 Powered-phase execution: measured versus assumed residual

The dispersion pipeline assumes a 25 m s^{-1} -per-axis departure residual (engineering assumption). This item measures what the powered phase actually leaks. A perturbed clone of the certified 3-D propagator flies the planar winner open-loop under a fractional thrust-magnitude bias $\varepsilon \sim N(0, 2.1 \times 10^{-3})$ (mapped from the assumed 50 m s^{-1} magnitude channel) plus pointing noise, with the heritage energy-trigger cutoff and perfect energy knowledge. The unperturbed control reproduces the carried $23.984510138157503 \text{ km s}^{-1}$ bitwise (verified). Table 10 summarizes.

The verdict is quantitative (measured). The assumed 25 m s^{-1} -per-axis residual is optimistic by a factor of 4 against the measured worst-component 99.6 m s^{-1} (e1), and at the assumed thrust-bias level 19 to 24% of open-loop runs miss the energy cutoff entirely. Both numbers are upper bounds on a real mission: this is open-loop execution of a fixed policy, a real mission observes a thrust-scale bias within days and re-optimizes, and the 50 m s^{-1} velocity-knowledge channel after telemetry-closed cutoff is not probed here. Section 6 carries both the assumed level and the measured level as explicit variants and chooses neither.

5.3 Linearity of the kinematic model: bounded

The entire crossing-epoch analysis assumes constant-velocity stellar motion. Closed forms bound the neglected physical terms (Table 11), including Proxima Centauri’s gravitational pull on the AB barycenter, a term the Fermi team identified, confirmed here by full Kepler-arc integration: 105.6 AU of track deflection at the crossing ($5.4\times$ the largest prior position row; cross-track 79.3 AU, 3.1% of the 2,600 AU success radius), worth +7.1 yr on the crossing, inside the Fermi team’s own 2 to 12 yr estimate, and 0.0030 km s^{-1} on v_∞ . The model: masses $M_{AB} = 1.988$ and $M_P = 0.1221 M_\odot$, current separation $r_0 = 12,057 \text{ AU}$ from Gaia DR3 Proxima astrometry (Gaia Collaboration 2023, A&A 674, A1) with the Akeson parallaxes (more current than the $r \approx 12,947 \text{ AU}$ of Kervella, Thévenin & Lovis 2017), propagated as a bound Kepler arc ($a \approx 7,573 \text{ AU}$, period $\approx 454 \text{ kyr}$), evaluated at the 79,372 yr working-state crossing (the same artifact gives 107.3 AU at 80,000 yr, so the other crossing epochs move the term by under 2 AU). Evaluated at that $r = 12,947 \text{ AU}$, the constant-acceleration variant of our model returns 90.6 AU (the Kepler arc gives 105.6 at our r_0), consistent with their $\approx 92 \text{ AU}$ up to the input difference; the term is deterministic and removable by adding Proxima to the propagation. The combined worst case moves the crossing by 56.4 yr (0.071% of T_{cross}) and v_∞ by 0.0168 km s^{-1} , far below the Kervella–Akeson catalog spread (414 yr between the Kervella and Akeson working states; 319 yr on the epoch-consistent Akeson state) and the 1,459 yr one-sigma radial-velocity lever. The linearity claim holds (measured closed-form bounds).

6. Joint end-to-end Monte Carlo

The joint model couples, in one Monte Carlo, everything the preceding sections treated separately: catalog and aim uncertainty (outer loop, 10,000 draws per scenario, seeds 20260806+idx), execution dispersion (inner loop, 1,000 draws per outer draw, deterministic seed lattice), the pumped Δv table (best-found upper bounds plus the $\sigma_{\text{surr}} = 0.0091 \text{ km s}^{-1}$ roughness proxy of Section 3.3), and the closure-margin model of Section 5.1. The evidentiary label carries verbatim from the artifact: simulation, engineering-assumption execution levels, heuristic upper-bound pump surrogate (margins are lower bounds), joint P conditioned on all stated assumptions. Table 12 reports the four astrometric scenarios.

The findings rank cleanly (simulation). The headline joint probability at the Kervella crossing is $P = 0.964$ at the assumed execution levels; the ephemeris-off control gives 0.986, so ephemeris uncertainty costs 2.2 points and execution the rest. Radial velocity dominates everything upstream: 99.1% of crossing-epoch variance and 64.5% of margin variance (one-at-a-time normalization), with the basin surrogate at 2.9% of margin variance. Stressing with the measured powered-phase residual drops P to 0.606, and multiplying by the open-loop departure bracket ($1 - 0.2355$) gives 0.737 (assumed) and 0.463 (stressed), both conservative brackets, since the failure mode is a thrust-scale bias a real mission would observe and re-optimize away. The RV convention of Section 4.5 moves the central epoch 299 yr (epoch-consistent Akeson) to 332 yr (Kervella; 349 yr on the working-state pair of Table 12) and changes joint P by less than 0.001: the design is insensitive to the convention. LEO-150’s joint probability never exceeds 0.05 in any scenario of Table 12, consistent with its unchanged closure verdict. No probability in this memorandum is scored at a fixed instant: both the dispersion pipeline and the joint model score closest approach (verified in source, with bit-exact draw replication).

Table 10. Powered-phase execution Monte Carlo (measured; seeds 20260804–20260807 by experiment). “Not reached” counts runs failing the energy target by the 12.0 yr wall (nominal burn 11.78 yr); survivor statistics are selection-biased and the deterministic ε sweep brackets the cliff between +0.0010 (reached) and +0.0020 (not reached, terminal energy 253 vs 280 km² s⁻²). e4 is the coarse-step control; its magnitude column is cutoff-quantization leakage, so e1 is the quoted measurement.

Exp	Pointing channel	n	dt	Not reached	$\sigma(\text{in-plane, m s}^{-1})$	$\sigma(\text{mag, m s}^{-1})$
e1	per-step noise	2,000	$\times 0.25$	23.6%	99.6	4.4
e2	run-coherent bias	1,000	$\times 0.25$	19.0%	121.8	4.3
e3	none (bias only)	1,000	$\times 0.25$	22.1%	96.5	4.4
e4	per-step noise	400	$\times 1.0$	38.3%	67.9	16.3

Table 11. Linearity bounds on the crossing epoch (measured closed forms). Perspective terms are exactly captured by Cartesian constant-velocity propagation; the quoted term is pure catalog-epoch bookkeeping (26 yr offset case; the 10 yr case is 11.7 yr). The wobble row is a bounded aim offset (19.4 AU maximum, 0.7% of the success radius); the 4.74 km s⁻¹ hazard of a non-barycentric proper motion does not apply because the Kervella 2016 solution is barycentric. Doppler factor 0.99992529. Displayed rows are rounded; the combined row sums the unrounded magnitudes.

Term	$ \Delta T_{\text{cross}} $ (yr)	$ \Delta v_{\infty} $ (km s ⁻¹)
Perspective acceleration (26-yr epoch offset)	30.36	0.0107
AB barycenter wobble (aim-offset bound)	8.28	0.0012
Proxima pull on AB barycenter (Kepler arc; Fermi-team term)	7.07	0.0030
Light-time / Doppler retardation	5.96	0.0018
Light-time retardation, catalog-epoch bound	4.37	–
Galactic tide over T_{cross}	0.33	0.0001
Combined worst case (sum of magnitudes)	56.36	0.0168

Table 12. Joint end-to-end Monte Carlo (simulation). Columns: central crossing epoch, central GTO-100 margin, joint P under assumed execution levels (ephemeris uncertainty on), joint P with the measured 99.6 m s⁻¹ powered-phase residual of Section 5.2 applied isotropically (conservative), and joint LEO-150 P at assumed levels. The kinematic rows carry the Section 4.5 RV correction. Scoring convention: the inner dispersion model scores the miss at closest approach t_{ca} (T_{nom} sets the aim state only); the two Akesson-based rows run on the working state of Table 5, with the J2019.5-state values in the text.

Scenario	T_{cross} (yr)	Margin (pts)	P (assumed)	P (meas. resid.)	P LEO-150
Kervella, carried RV	79,785.7	+3.82	0.964	0.606	0.046
Akesson, carried RV	79,371.9	+3.73	0.984	0.629	0.000
Kervella, kinematic RV	80,117.9	+3.84	0.964	0.603	0.014
Akesson, kinematic RV	79,720.8	+3.74	0.983	0.625	0.000

Radial-velocity error is along-track (correlation with miss +0.010), moving the crossing date while leaving the miss almost unchanged. On the Akesson J2019.5 catalog state the central dispersion probability is 0.9858 (kinematic) / 0.9866 (observed- γ), and the joint P of Table 12 sits 0.0010 to 0.0011 below the central dispersion P in every scenario, so the J2019.5-state joint P is ≈ 0.985 to 0.986; full regeneration of the joint table and sensitivity artifacts on the J2019.5 state is the single recommended follow-up work order (the golden fixture of Section 7.3, pinned to that state, accompanies this memorandum).

7. Recommendation

7.1 The publishable arrival statement: two exact wording options

Option A is the recommended primary wording, built to the adopted program convention: the Akeson 2021 state pinned at its native epoch J2019.5, the kinematic radial-velocity frame, and T counted from the 2029.0 departure, with the Fermi team's ± 400 –800 yr band estimate adopted as a program input.

Option A (recommended; Akeson J2019.5 state, kinematic frame, T from departure 2029.0). “The required departure asymptote crosses the ecliptic plane at $T = 79,756$ yr after the 2029.0 departure, crossing date AD 81,785, program band ± 400 –800 yr (measured; Akeson et al. 2021 barycentric state at its native epoch J2019.5, kinematic frame -22.3182 ± 0.0034 km s $^{-1}$, frame correction per Kervella, Thévenin & Lovis 2017 (A&A 598, L7), central value 79,756.4 yr, crossing distance 6.428 ly, v_∞ 24.158 km s $^{-1}$). The observed- γ variant ($V_0 = -22.3796$ km s $^{-1}$ with the 0.10 km s $^{-1}$ carried allowance) gives 79,457.5 yr, AD 81,486.5; the frame split is 298.9 yr at the measured 4,870 yr-per-km s $^{-1}$ slope. The catalog-frame scenario spread (Akeson/Kervella $\times$ observed/kinematic) spans AD 81,486 to 82,118, half-spread ≈ 316 yr, supporting the band floor; the cross-catalog proper-motion systematic and the Proxima-pull term argue against narrowing below ± 400 yr, and the Fermi team's published 79,252 yr lies inside the band ceiling under either origin bookkeeping. The breaking point is the radial-velocity uncertainty; a 0.03 km s $^{-1}$ kinematic measurement would narrow the band to ± 200 yr.”

Option B fits a document that must stay Kervella-primary for consistency with the certified feasibility chain.

Option B (Kervella-primary alternative). “The required departure asymptote crosses the ecliptic plane at $T = 79,786 \pm 1,470$ yr on the Kervella state clock (measured; Kervella et al. 2016 catalog parallax, proper motions, and gamma end to end on a J2000-family state basis, spectroscopic frame, diagonal covariance); the kinematic-frame central value is 80,118 yr, a $+0.23\sigma$ shift. The epoch-consistent Akeson et al. 2021 state gives 79,467.0 yr from J2019.5 (spectroscopic V_0) and 79,765.9 yr (kinematic), and the Fermi Explorer team's independent analysis gives 79,252 yr; all values agree within one sigma of either band. The breaking point is the radial-velocity uncertainty, at 4,860 yr of crossing-epoch shift per km s $^{-1}$, with the frame convention worth a further one-sided 330 to 350 yr.”

One bookkeeping caution on Option B: the carried Kervella state pairs the K16 catalog observables with a project position constant on a J2000-family basis (the K16 catalog barycenter is published at epoch J1991.25; Table 5), so its T is counted on that state clock, and it is not counted from the 2029.0 departure. The epoch-consistency analysis of Section 4.4 covers the Akeson state only; we did not apply it to the Kervella state, which is one more reason Option A is the recommended wording.

The rationale for Akeson 2021 as the primary basis: the most recent catalog with the smallest published uncertainties (V_0 sigma 0.0020 km s $^{-1}$; the 0.10 km s $^{-1}$ the variant carries is an adopted absolute-accuracy allowance for frame and zero-point systematics that appears in no catalog) and a native J2019.5 epoch that the exchange state pins. The rationale for the kinematic frame: it is the true line-of-sight motion the trajectory needs, and the spectroscopic value hides the verified -61.4 ± 2.7 m s $^{-1}$ differential gravitational-redshift bias, whose $+61.4$ m s $^{-1}$ correction is worth $+299$ yr on the epoch-consistent Akeson base (Section 4.5). The quoted band is a program input adopted from the Fermi team's estimate (it is not a propagated covariance); the Akeson-based carried-sigma width (± 489) falls inside its range, the Kervella width ($\pm 1,468$) exceeds it, and the 316 yr scenario half-spread of Option A sits below its floor. The covariance is assumed diagonal throughout.

7.2 Joint action list for the Fermi team

- **Adopt one catalog end to end, at its native epoch.** We recommend Akeson 2021 pinned to J2019.5 for both teams: barycenter $219.85892215^\circ/-60.83163195^\circ$, parallax 750.81 ± 0.38 mas, proper motion $(-3639.95 \pm 0.42, +700.40 \pm 0.17)$ mas yr $^{-1}$, $V_0 = -22.3796 \pm 0.0020$ km s $^{-1}$ spectroscopic. This removes the entire 533.5 yr input effect and the 0.094 km s $^{-1}$ raw v_∞ offset at once.
- **State the radial-velocity convention and adopt the kinematic frame.** Their `astro.py` carries the spectroscopic -22.4 km s $^{-1}$ for AB with no stated convention; one comment line fixes it. (The matching housekeeping item on our side is the mixed-convention constants module of Section 4.5; their repository is clean.) The kinematic AB values are -22.332 km s $^{-1}$ (Kervella base) and -22.318 km s $^{-1}$ (Akeson base); the correction moves the crossing 299 to 332 yr later and changes no verdict (measured).
- **Exchange one state vector.** A single ecliptic-frame position and velocity at the pinned epoch J2019.5 (program input, accepted), printed to full precision, verifies the whole kinematic chain in one comparison.

Our reproduction matrix shows conventions contribute 0.01 yr; the exchange makes that check permanent, and the golden fixture below automates it.

- **Treat the radial velocity as the program risk.** It carries 4,864 yr per km s^{-1} of crossing-epoch shift (measured), 16 times the parallax leverage under the Kervella-based carried allowances and 9 times under the Akeson-based, it drives 99.1% of the joint arrival-epoch variance (simulation), and its convention alone is worth 330 to 350 yr across the Kervella and Akeson working bases (measured; Table 7). Any future refinement effort should buy kinematic radial-velocity precision first.
- **Keep the crossing epoch as the design default.** Their design argument holds and this analysis priced it: the crossing is fixed by geometry, costs 22 to 27 m s^{-1} over the measured basin bottom (measured; 0.11% of total), keeps arrival probability between 0.965 and 0.987 (simulation; 0.965 Kervella, 0.984 Akeson working state, 0.9858/0.9866 Akeson J2019.5 state), closes GTO-100 with +3.8 pts margin (simulation), and moved by zero when their pricing model was corrected. Quote it with the astrometric band above; the band is the honest content of the number.

7.3 Offer: a golden fixture for permanent cross-validation

We offer the Fermi team a golden fixture that makes the code-versus-inputs separation permanent. The fixture is pinned to the exchange state of this memorandum: the Akeson 2021 barycenter observables at their native epoch J2019.5, used epoch-consistently, in the kinematic radial-velocity frame ($-22.3182 \pm 0.0034 \text{ km s}^{-1}$), with the state clock at J2019.5 and the departure-2029.0 origin reported alongside. The package is four small files: the agreed astrometric state with pinned unit constants and five test epochs; the expected outputs, $T_{\text{cross}} = 79,765.87 \text{ yr}$ on the state clock (79,756.37 yr from departure, crossing date AD 81,785.37), crossing distance 6.427805 ly, v_{∞} at the crossing 24.15829 km s^{-1} , and v_{∞}/β at the five epochs, fifteen checked values in all, with stated tolerances of 0.02 yr, 10^{-4} (km s^{-1} and degrees), and 10^{-6} ly; a standalone stdlib-only Python checker that recomputes and asserts every value; and a README declaring every convention (state epoch, time origins, frame, and the observed- γ variant). The checker passes on a clean system interpreter with no project code on the path (verified), and an independent recompute through a different call path agrees to machine precision. If the Fermi implementation reproduces the expected outputs, any future disagreement between the two teams is attributable to inputs alone, with code ruled out; one run replaces an entire reconciliation exercise. The package accompanies this memorandum.

7.4 Verdict summary

The 534 yr gap is an input difference: distance carries 528.8 yr of it, the convention effect is -0.01 yr , and the gap is not statistically significant (the Fermi value sits at -0.36σ in the Kervella band and -0.25σ in the Akeson band; the ours-minus-Fermi separation itself is 1.09σ of the Akeson band, still inside statistical agreement). Their 79,252 yr reproduces exactly from their inputs (verified). Their 77,500 yr PP optimum agrees with our independent model to 1 yr (verified) and sits 16 m s^{-1} above the measured basin bottom, inside a basin flat to 25 m s^{-1} (measured). The 73,012 yr impulsive optimum stands (verified). The measurement program returned six confirmed verdicts from six items, with every anchor gate passing at bit-level or better and no tolerance tuned to pass, and the one number it adds to the headline set is the joint $P = 0.964$ (simulation; assumed execution levels) for the recommended crossing-epoch design. No tuning was performed on either implementation at any point.

A. Methods, seeds, and reproducibility

This appendix is self-contained: it states every input, gate, and seed needed to reproduce the memorandum's numbers from public information, with no reference to internal file layouts.

A.1 Inputs

Two Alpha Centauri AB barycentric states, ICRS, with linear motion: *Kervella 2016 basis*: catalog parallax $747.17 \pm 0.61 \text{ mas}$, $\mu_{\alpha*} -3619.9$ and $\mu_{\delta} +693.8 \text{ mas yr}^{-1}$ (± 3.9 each, catalog sigmas), catalog gamma $-22.3930 \pm 0.0043 \text{ km s}^{-1}$ carried as -22.4 with a 0.30 km s^{-1} carried working sigma (adopted allowance), all from A&A 594, A107 (primary solution). The position RA 219.90206°, Dec -60.83399° paired with these observables is

a project constant on a J2000-family basis, and it is not a K16 catalog value (the K16 catalog barycenter is published at epoch J1991.25), so the carried Kervella state runs on a J2000-family state clock. *Akeson 2021* (AJ 162, 14, native epoch J2019.5): barycenter $219.85892215^\circ/-60.83163195^\circ$, parallax 750.81 ± 0.38 mas, $\mu_{\alpha*} -3639.95 \pm 0.42$ and $\mu_\delta +700.40 \pm 0.17$ mas yr $^{-1}$, $V_0 = -22.3796 \pm 0.0020$ km s $^{-1}$. The Akeson-based working state pairs the Akeson parallax with adopted proper motions ($-3644.4, +702.8; \pm 1.5$ mas yr $^{-1}$ working sigmas; no catalog provenance) and the K16/PB16 gamma rounded to -22.39 with a 0.10 km s $^{-1}$ allowance on a J2000-family position; the working-state Monte Carlo rows of the main text run on it (Table 5). The Fermi team's published inputs, from their public repository: RA 219.9021° , Dec -60.834° , distance 4.344 ly, $\mu_{\alpha*} -3620.0$ and $\mu_\delta +694.0$ mas yr $^{-1}$, radial velocity -22.4 km s $^{-1}$. Radial-velocity source values for the convention audit (verified against re-fetched source text): Kervella 2016 orbit gamma -22.393 km s $^{-1}$, Pourbaix–Boffin 2016 (A&A 586, A90) -22.390 km s $^{-1}$, Akeson 2021 $V_0 = -22.3796$ km s $^{-1}$, all spectroscopic; the kinematic correction adds $+61.4 \pm 2.7$ m s $^{-1}$, removing the differential gravitational redshift $\Delta v_{\text{GRS}} = -61.4 \pm 2.7$ m s $^{-1}$ published by Kervella, Thévenin & Lovis 2017 (A&A 598, L7).

A.2 Crossing-epoch computation

Rotate the state to ecliptic coordinates with J2000 mean obliquity 23.43928° . Convert proper motion with $\kappa = 4.74047$ km s $^{-1}$ per arcsec yr $^{-1}$ at 1 pc and a Julian year of 31,557,600 s. Propagate linearly and solve $\beta(T) = 0$ for the required asymptote, equivalently $T = -z_0/v_z$. The convention matrix of Section 4 swaps the obliquity digits (23.439281), the light-year value (9,460,730,472,580.8 km against 9.46073×10^{12} km), and the κ digits (4.740470463) between the two sides; the combined effect on the crossing epoch is 0.01 yr (measured).

A.3 Astrometric Monte Carlo band

10,000 draws, seed 20260803, independent Gaussians on parallax, both proper-motion components, and radial velocity (diagonal covariance, assumed; no published cross-terms are carried in either input set). The linear one-sigma propagation agrees with the Monte Carlo sigma within 0.5% for both catalogs (measured). The full percentile sets (measured, in years, spectroscopic frame):

- **Kervella 2016 state:** 2.5% 77,029; 16% 78,351; 50% 79,793; 84% 81,287; 97.5% 82,755; mean 79,816; Monte Carlo sigma 1,462; linear sigma 1,468.
- **Akeson-based working state:** 2.5% 78,434; 16% 78,888; 50% 79,374; 84% 79,862; 97.5% 80,332; mean 79,376; Monte Carlo sigma 487; linear sigma 489.

The kinematic-frame values shift each percentile later by 332 yr (Kervella) or 349 yr (Akeson working base; 299 on the epoch-consistent state); the 2.7 m s $^{-1}$ correction uncertainty adds 13 yr, inside the stated sigmas (computed).

A.4 3-D tilt measurements

The tilt table uses the certified 3-D pump propagator of the feasibility chain with a 2,600 AU miss credit and the AB barycenter aim. Acceptance per row: the six heritage gates plus a direction gate $|\beta_{\text{achieved}} - \beta_{\text{target}}| \leq 0.04^\circ$; at the crossing epoch the gate reads $|\beta_{\text{achieved}}| \leq 0.04^\circ$. Each row runs a time-step ladder at $\times 1.0/\times 0.5/\times 0.25$ and carries the maximum gate-passing value; ladder spread per row is under 0.4%. Search recipe: 600 coarse plus 8 polish starts (3-D) and 400 plus 6 (planar), Nelder–Mead polish with one deterministic restart, seed base 20260803 with fixed per-row offsets. Anchors: the planar embedding reproduces the heritage planar value to 10^{-9} km s $^{-1}$, and the certified 3-D ladder reproduces the heritage 24.5627 km s $^{-1}$ anchor exactly (verified). The 3-D totals are best-found upper bounds with ± 0.2 km s $^{-1}$ seed scatter; the tilt increments in Table 1 are differences of matched runs under identical seeds.

A.5 Basin sweep and adjudication

The empirical basin sweep runs epochs 74,000 to 81,000 yr in 500 yr steps, five seeds per epoch, seed base 20260804 with explicit per-run offsets, and the bounded 3-D search recipe imported verbatim from the certified heritage code. The four anchor gates of Table 2 pass before any sweep value is read. The sweep com-

pleted in 1,715 s against a 2,100 s budget with all 75 cells feasible at 5/5 seeds. The adjudication run re-ran the 79,500 yr cell with five fresh seeds and the 80,500 yr cell with three, spot re-ran 78,500 and 79,000 (both exact reproductions), and re-laddered both outlier winners on an extended ladder down to $\times 0.0625$. The surrogate-roughness check compares the joint Monte Carlo's $\sigma_{\text{surr}} = 0.0091 \text{ km s}^{-1}$ against the measured quadratic-detrend rms 0.0073 km s^{-1} of the adjudicated envelope.

A.6 Dispersion and closure at the design epochs

The dispersion runs use the certified pipeline with 20,000 draws per case and the published error budget: pointing 0.10° per axis, magnitude 50 m s^{-1} , residual 25 m s^{-1} per axis (all engineering assumptions), carried ephemeris sigmas (Table 5), and the 2,600 AU success radius. Seeds: 42 (published baseline anchor, reproduced exactly), 20260804 (Kervella crossing), 20260805 (Akeson crossing), 20260806 (catalog control). Closure interpolates the 12-yr pumped $\Delta\nu$ column between the 79,000 and 80,000 yr best-found rows and extrapolates the bracket-end margins at the measured slopes -1.818 (GTO-100) and -1.627 (LEO-150) points per km s^{-1} .

A.7 Powered-phase execution Monte Carlo

A perturbed clone of the certified 3-D propagator flies the planar winner open-loop under a fractional thrust-magnitude bias $\varepsilon \sim N(0, 2.1 \times 10^{-3})$ plus the pointing channels of Table 10, with the heritage energy-trigger cutoff, a 12.0 yr wall against the 11.78 yr nominal burn, and perfect energy knowledge. Seeds 20260804 to 20260807 by experiment. The unperturbed control reproduces the carried $23.984510138157503 \text{ km s}^{-1}$ bitwise (verified). The deterministic ε sweep brackets the completion cliff between $+0.0010$ and $+0.0020$.

A.8 Joint Monte Carlo

Outer loop: 10,000 astrometric draws per scenario, seeds 20260806+idx. Inner loop: 1,000 execution draws per outer draw on a deterministic seed lattice. The pumped $\Delta\nu$ surrogate carries the best-found table plus $\sigma_{\text{surr}} = 0.0091 \text{ km s}^{-1}$; margins inherit the closure model above. Variance attributions use one-at-a-time normalization. The ephemeris-off control (0.986) isolates the ephemeris cost at the Kervella crossing.

A.9 Honesty constraints and the adversarial pass

Neither implementation was adjusted toward the other at any point. The reproduction of the Fermi crossing epoch ran their published inputs through a clean reimplementation, and the match (0.25 yr, inside their integer rounding) required no fitting. The quadratic tilt model was fixed from a single prior anchor before the multi-epoch measurement existed. All stochastic results carry the fixed seeds stated in this appendix, and negative controls (the planar-collapse residual, 0.015 km s^{-1}) are reported alongside the positive results. An independent adversarial verification pass re-ran all analysis scripts to completion (artifacts bit-identical up to wall-clock strings), recomputed headline values through different call paths, and confirmed all six verdicts. No tolerance was found tuned to pass: all anchor gates are bit-level, 10^{-6} , or 10^{-9} , and the one loose gate (the 0.05 km s^{-1} planar-collapse gate against a measured 0.0149 km s^{-1}) is declared in-script with its prior measured value referenced.